

Proofs

Proofs are the primary means by which mathematical truth is established. The purpose of a proof is to justify the truth of a mathematical statement (called the **conclusion**) by considering other mathematical statements (called **premises**). Once a statement is proved, we refer to it as a **theorem**.

- **Definition 1: Conjunctions ($P \wedge Q$)**

A statement of the form $P \wedge Q$ is true when both P and Q are true. To prove a conjunction, it suffices to prove each of its component statements separately. That is, if we establish P and we also establish Q , we may conclude $P \wedge Q$.

- **Example 1.1**

To prove that a number n is even and greater than 10, we prove separately that n is an even number and that $n > 10$. Combining them yields the result.

- **Definition 2: Disjunctions ($P \vee Q$)**

A statement of the form $P \vee Q$ is true whenever at least one of its components is true. Therefore, to prove a disjunction, it is enough to prove one of its disjuncts (though a false disjunct cannot establish the disjunction).

- **Example 2.1**

To prove that a number n is either even or odd, it suffices to prove that n is even, and the disjunction follows.

- **Example 2.2**

Let $a, b \in \mathbb{R}$ that satisfy $a + b = 10$. Prove that $ab \leq 25$ or $a^2 + b^2 < 10$.

- **Solution 2.2.1**

Suppose $a, b \in \mathbb{R}$ satisfy $a + b = 10$, then $a = 10 - b$, and we have

$$\begin{aligned} ab &= (10 - b)b \\ &= 10b - b^2 \\ &= -(b^2 - 10b) \\ &= -(b^2 - 10b + 25) + 25 \\ &= -(b - 5)^2 + 25 \\ &\leq 25 \end{aligned}$$

So $ab \leq 25$, then $ab \leq 25$ or $a^2 + b^2 < 10$.

- **Example 2.3**

Prove that $(\forall x) P(x) \vee (\forall x) Q(x) \implies (\forall x) (P(x) \vee Q(x))$ and refute the converse implication.

- **Solution 2.3.1**

Suppose $(\forall x) P(x) \vee (\forall x) Q(x)$ is true, then $(\forall x) P(x)$ is true or $(\forall x) Q(x)$ is true. Suppose $(\forall x) P(x)$ is true, then $\forall x, P(x)$ is true, then $\forall x, P(x) \vee Q(x)$ is true and then $\forall x, (P(x) \vee Q(x))$ is true.

To check that the converse is false, consider $P(x) : x$ is even, $Q(x) : x$ is odd.

- **Definition 3: Indirect Strategy for Disjunction**

Another useful strategy to prove $P \vee Q$ is to proceed indirectly. Assume that one of the disjuncts is false (say $\neg P$) and then prove the other (Q):

$$P \vee Q \equiv \neg P \implies Q$$

- **Definition 4: Proof by Cases**

If we know that $P \vee Q$ is true, and we can prove R from P , and also prove R from Q , we may conclude R :

- **Example 4.1**

Suppose a natural number is either even or odd, and we want to prove that n^2 is even. We divide the argument into two cases: if n is even, n^2 is even; if n is odd, n^2 is even. Since one of these cases must be true, n^2 is even in either case.

- **Example 4.2**

Proof: $P \wedge (Q \vee R) \equiv (P \wedge Q) \vee (P \wedge R)$

- **Solution 4.2.1**

We want to prove $\forall a, b, c \in \mathbb{R}, \min\{a, \max\{b, c\}\} = \max\{\min\{a, b\}, \min\{a, c\}\}$. Let

$$M_1 := \min\{a, \max\{b, c\}\}$$

$$M_2 := \max\{\min\{a, b\}, \min\{a, c\}\}$$

we want to prove $M_1 \leq M_2$ and $M_2 \leq M_1$.

For the case $M_1 \leq M_2$, by the definition of $\min$ and $\max$

$$\begin{aligned} M_1 &\leq a \wedge M_1 \leq \max\{b, c\} \\ \implies M_1 &\leq a \wedge (M_1 \leq b \vee M_1 \leq c) \\ \implies (M_1 &\leq a \wedge M_1 \leq b) \vee (M_1 \leq a \wedge M_1 \leq c) \\ \implies M_1 &\leq \min\{a, b\} \vee M_1 \leq \min\{a, c\} \\ \implies M_1 &\leq \max\{\min\{a, b\}, \min\{a, c\}\} \\ \implies M_1 &\leq M_2 \end{aligned}$$

For the case $M_2 \leq M_1$, by the definition of $\min$ and $\max$, we have

$$\min\{a, b\} \leq a, \min\{a, b\} \leq b \leq \max\{b, c\}$$

then $\min\{a, b\} \leq \min\{a, \max\{b, c\}\}$.

The other inequality is almost the same. Then $M_2 = \max\{\min\{a, b\}, \min\{a, c\}\}$.

- **Solution 4.2.2**

We want to prove $\forall a, b, c \in \mathbb{R}, \min\{a, \max\{b, c\}\} = \max\{\min\{a, b\}, \min\{a, c\}\}$.

Suppose $a, b, c \in \{0, 1\}$. We split into two cases depending on the value of a .

Case 1 $a = 0$

$$\begin{aligned} \text{LHS} &= \min\{0, \max\{b, c\}\} = 0 \\ \text{RHS} &= \max\{\min\{0, b\}, \min\{0, c\}\} = \max\{0, 0\} = 0 \end{aligned}$$

Thus LHS = RHS.

Case 2 $a = 1$

$$\begin{aligned} \text{LHS} &= \min\{1, \max\{b, c\}\} = \max\{b, c\} \\ \text{RHS} &= \max\{\min\{1, b\}, \min\{1, c\}\} = \min\{b, c\} \end{aligned}$$

Thus LHS = RHS.

Since equality holds for $a = 0$ and $a = 1$, the identity is true for all $a, b, c \in \{0, 1\}$.

- **Example 4.3**

Let $a_1, a_2, \dots, a_7 \in \mathbb{Z}$, prove that among these seven integers, one can always choose four numbers whose sum is even.

- **Solution 4.3.1**

Suppose $a_1, a_2, \dots, a_7 \in \mathbb{Z}$. Let K be the number of odd integers among $a_1, a_2, \dots, a_7$.

If $4 \leq K \leq 7$, then take 4 of these odd integers, their sum is even:

$$(2a + 1) + (2b + 1) + (2c + 1) + (2d + 1) = 2(a + b + c + d + 2)$$

If $0 \leq K < 4$, then take 4 of these even integers, their sum is even:

$$2a + 2b + 2c + 2d = 2(a + b + c + d)$$

In both cases, we obtain that we can always choose four numbers whose sum is even.

◦ **Example 4.4**

Prove that $\forall n \in \mathbb{Z}$, if n is odd, then $3n$ is odd.

■ **Solution 4.4.1**

Suppose $n \in \mathbb{Z}$, and n is odd. Then, there exists an integer $k \in \mathbb{Z}$ such that $n = 2k + 1$. Then

$$\begin{aligned} 3n &= 3(2k + 1) \\ &= 6k + 3 \\ &= 2(3k + 1) + 1 \end{aligned}$$

Since $3k + 1$ is an integer, $3n$ is odd.

◦ **Example 4.5**

For every $a, b, c \in \mathbb{Z}$, prove that if $a \mid b$ and $b \mid c$, then $a \mid c$.

■ **Solution 4.5.1**

Suppose $a, b, c \in \mathbb{Z}$ and $a \mid b \wedge b \mid c$. Then there exist $k, q \in \mathbb{Z}$ such that $b = ka$ and $c = qb$. Then $c = qka$, and since $qk \in \mathbb{Z}$, by definition, $a \mid c$.

◦ **Example 4.6**

Let $n \in \mathbb{Z}$. Prove that if n^2 is even, then n is even.

■ **Solution 4.6.1**

We will prove the contrapositive: $\forall n \in \mathbb{Z}$, if n is odd, then n^2 is odd.

Suppose $n \in \mathbb{Z}$ is odd. Then $\exists k \in \mathbb{Z}$ such that $n = 2k + 1$. Then

$$n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1$$

is odd because $2k^2 + 2k$ is an integer.

◦ **Example 4.7**

Prove that there is no smallest positive rational number.

■ **Solution 4.7.1**

By contradiction, suppose there exists a smallest positive rational number r . Then $\frac{r}{2}$ is a smaller positive rational number. This is a contradiction because r was assumed to be the smallest. Thus, there doesn't exist a smallest positive rational number.

◦ **Example 4.8**

$\forall \epsilon > 0$, prove that $\exists N \in \mathbb{N}$ such that $\forall n \geq N$, $\frac{1}{n} < \epsilon$.

■ **Solution 4.8.1**

Suppose $\epsilon > 0$. We will first prove that $\exists N \in \mathbb{N}$ such that $\frac{1}{N} < \epsilon$.

The Archimedean Property states that for any real number $x \in \mathbb{R}$, there exists a natural number $N \in \mathbb{N}$ such that $N > x$. By setting $x = \frac{1}{\epsilon}$ in the Archimedean Property, we guarantee that $\exists N \in \mathbb{N}$ satisfying $N > \frac{1}{\epsilon}$, and then $\epsilon > \frac{1}{N}$. Then if $n \geq N$, $\frac{1}{N} \geq \frac{1}{n}$ and $\frac{1}{n} < \epsilon$.

• **Definition 5: Direct Strategy for Implications ($P \implies Q$)**

To prove a statement of the form $P \implies Q$, we must show that whenever P is true, Q is also true.

◦ **Example 5.1**

To prove that if n is even, then $n + 1$ is odd, we assume that n is even and then show that $n + 1$ is odd.

◦ **Modus Ponens**

If we establish both $P \implies Q$ and P , we can conclude Q .

• **Definition 6: Indirect Strategy (Proof by Contrapositive)**

Since $P \implies Q \equiv \neg Q \implies \neg P$, we can prove $\neg Q \implies \neg P$ instead of $P \implies Q$.

• **Definition 7: Proof by Contradiction**

To prove $P \implies Q$, suppose that P is true and Q is false ($P \wedge \neg Q$) and show that this assumption leads to a contradiction ($R \wedge \neg R$):

$$P \implies Q \equiv (P \wedge \neg Q) \implies (R \wedge \neg R)$$

We have:

P	Q	R	$\neg Q$	$P \wedge \neg Q$	$R \wedge \neg R$	$P \implies Q$	$(P \wedge \neg Q) \implies (R \wedge \neg R)$
1	1	1	0	0	0	1	1
1	1	0	0	0	0	1	1
1	0	1	1	1	0	0	0
1	0	0	1	1	0	0	0
0	1	1	0	0	0	1	1
0	1	0	0	0	0	1	1
0	0	1	1	0	0	1	1
0	0	0	1	0	0	1	1

• **Definition 8: Disproofs and Counterexamples**

A statement is disproved if we show a single instance in which it is false, known as a **counterexample**.

For a universally quantified statement $\forall x P(x)$, a single object x_0 for which $P(x_0)$ is false suffices to disprove the statement.

For an implication $P \implies Q$, a counterexample consists of a situation where P is true and Q is false.

◦ **Example 8.1**

Consider the statement $x^2 - 1 > 0$ for all $x \in \mathbb{R}$. To disprove it, use the counterexample $x_0 = 1$, which gives $1^2 - 1 = 0 \not> 0$. Thus, the statement is false.

◦ **Example 8.2**

Prove that the existentially quantified statement $(\exists x \in \mathbb{R}) \left(\frac{1}{x^2+1} > 1 \right)$ is false.

■ **Solution 8.2.1**

- We need to show that $\frac{1}{x^2+1} \leq 1$ for every $x \in \mathbb{R}$. Since $0 \leq x^2$ for all $x \in \mathbb{R}$, it follows that $1 \leq x^2 + 1$. Because $x^2 + 1 > 0$, taking the reciprocal reverses the inequality:

$$\frac{1}{x^2 + 1} \leq 1$$

for every $x \in \mathbb{R}$. Thus, the original statement is false.